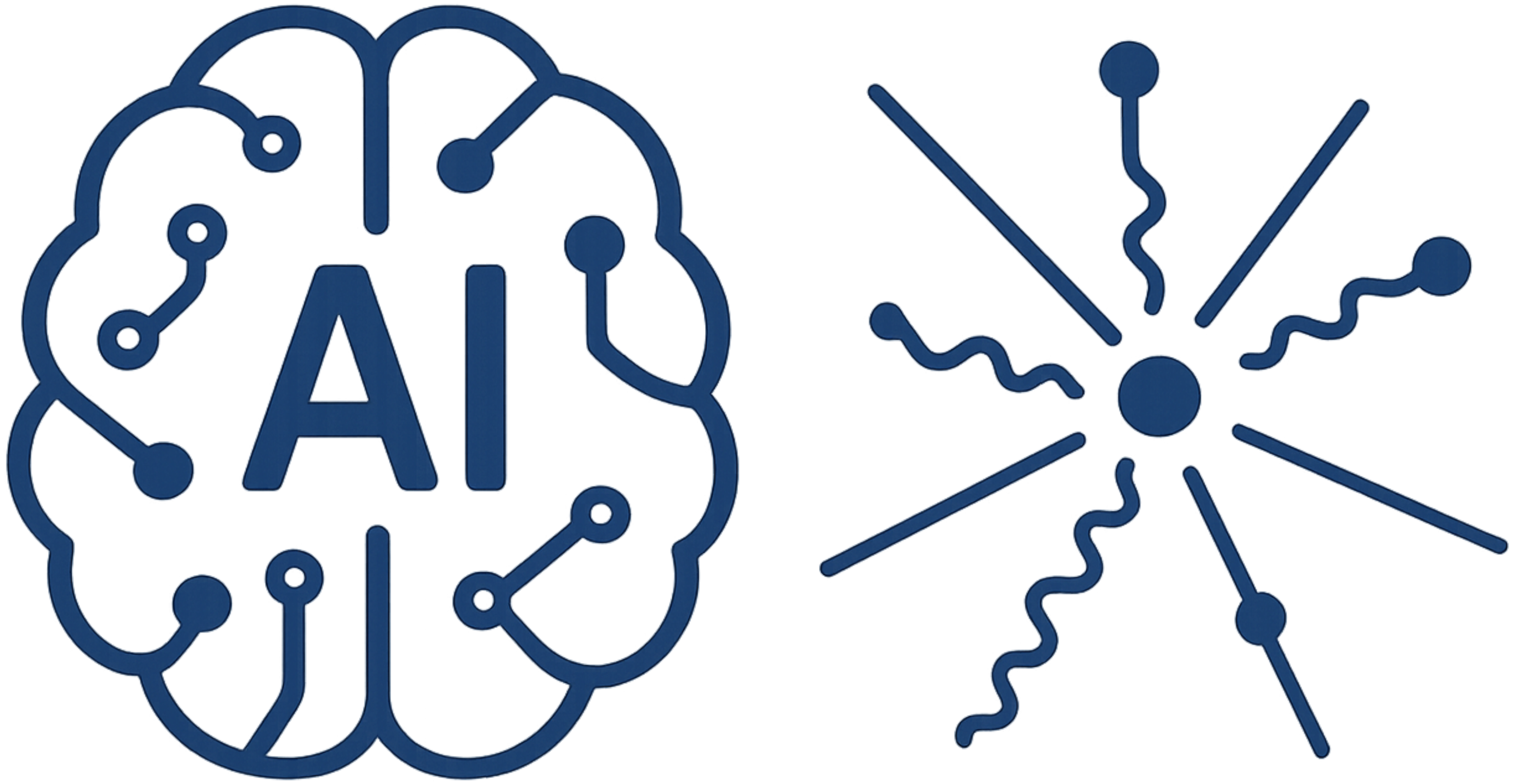
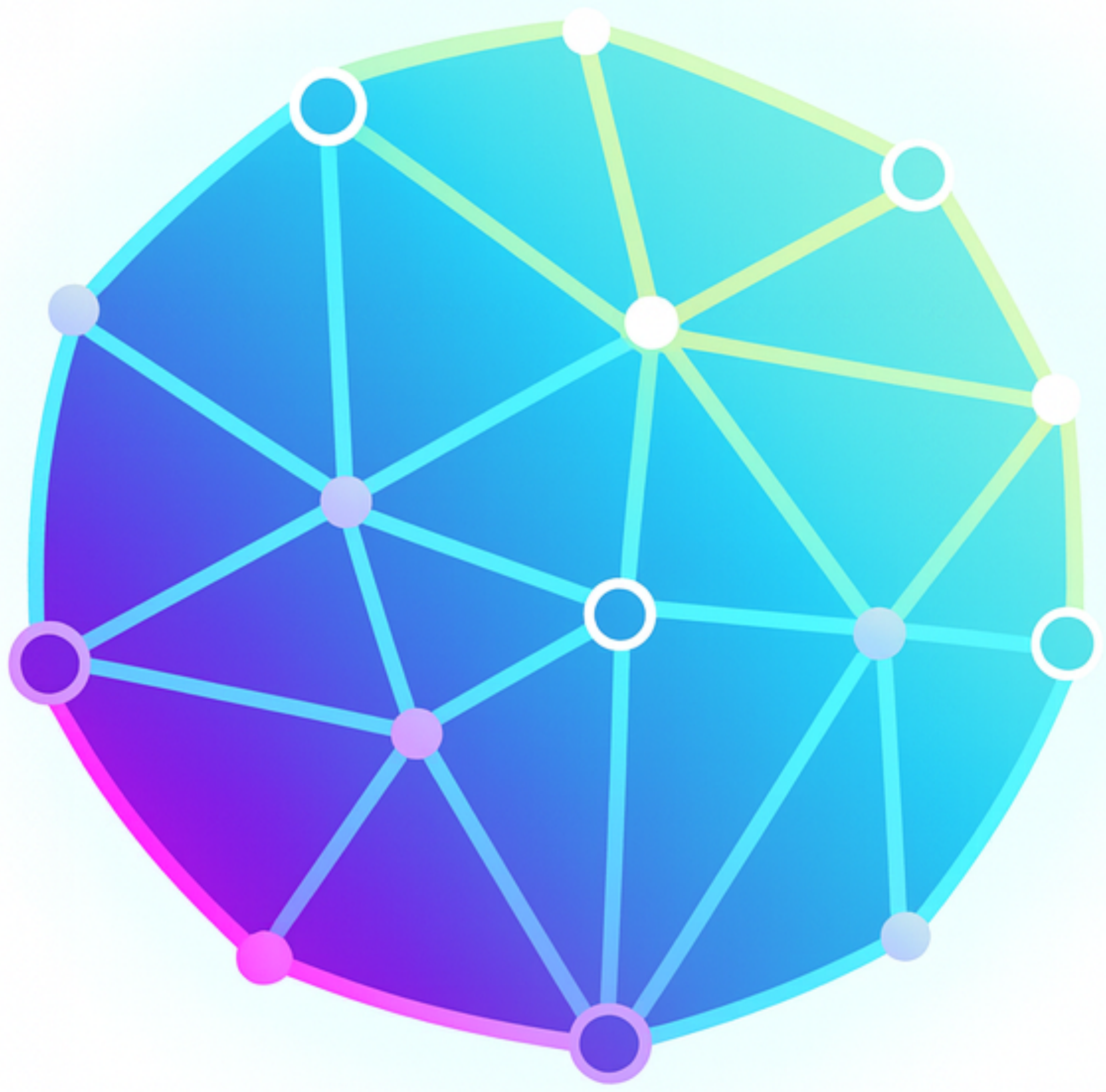
Introduction to

AI-Driven HEP



S. A. Fard - School of Physics (IPM)



Session 17

Foundation Models

Reference

[1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023.

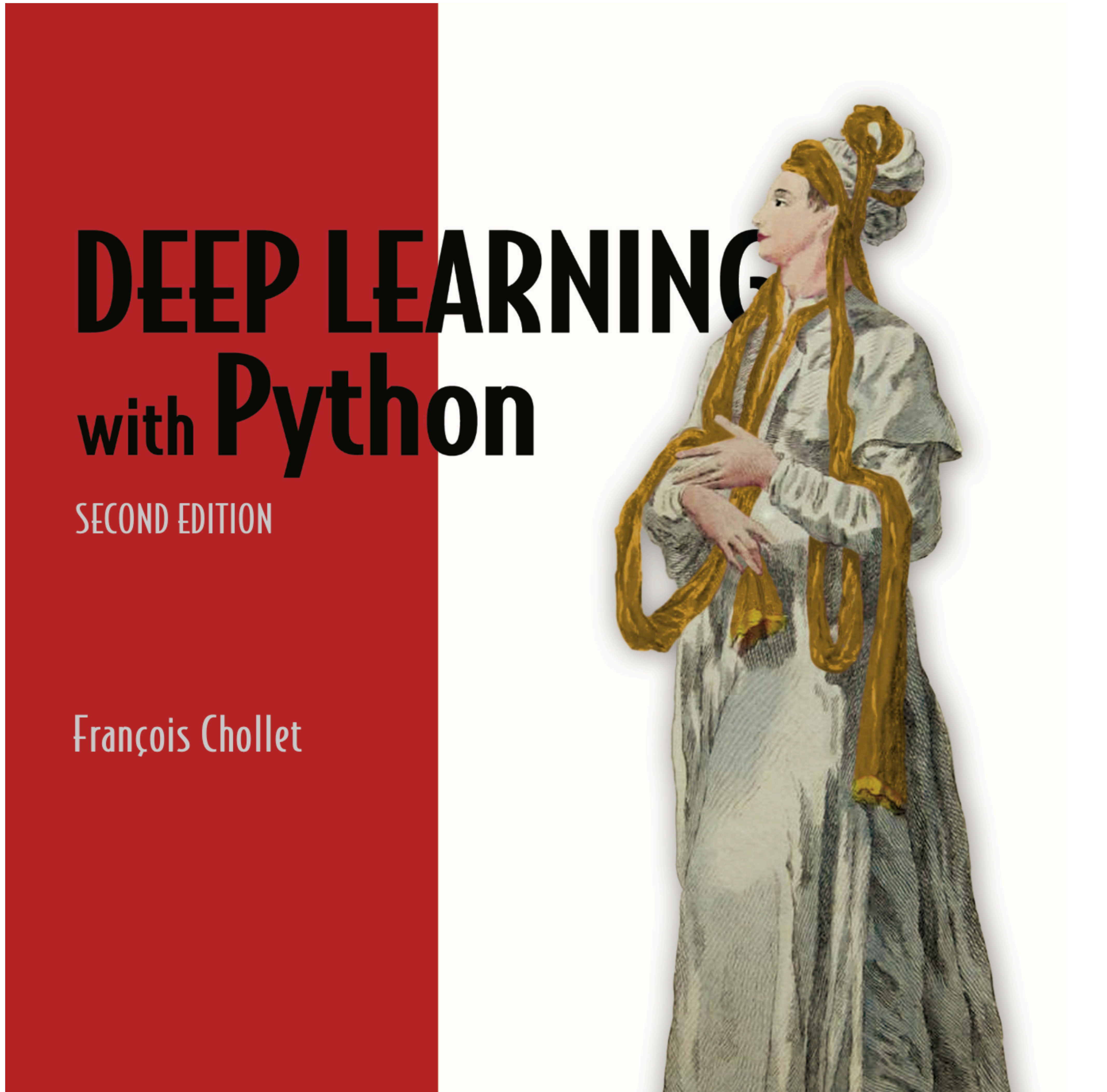
[2] Chollet, Francois. Deep learning with Python. (2021)

https://physics.ipm.ac.ir/~ansarifard/

پژوهشگاه‌دانش‌های‌بنیادی (مرکز تحقیقات فیزیک نظری و ریاضیات)

INSTITUTE FOR RESEARCH IN FUNDAMENTAL SCIENCES

“Habit of a Persian Lady in 1568.”, Thomas Jefferys, A Collection of the Dresses of Different Nations, Ancient and Modern (four volumes), London, published between 1757 and 1772



Transformer

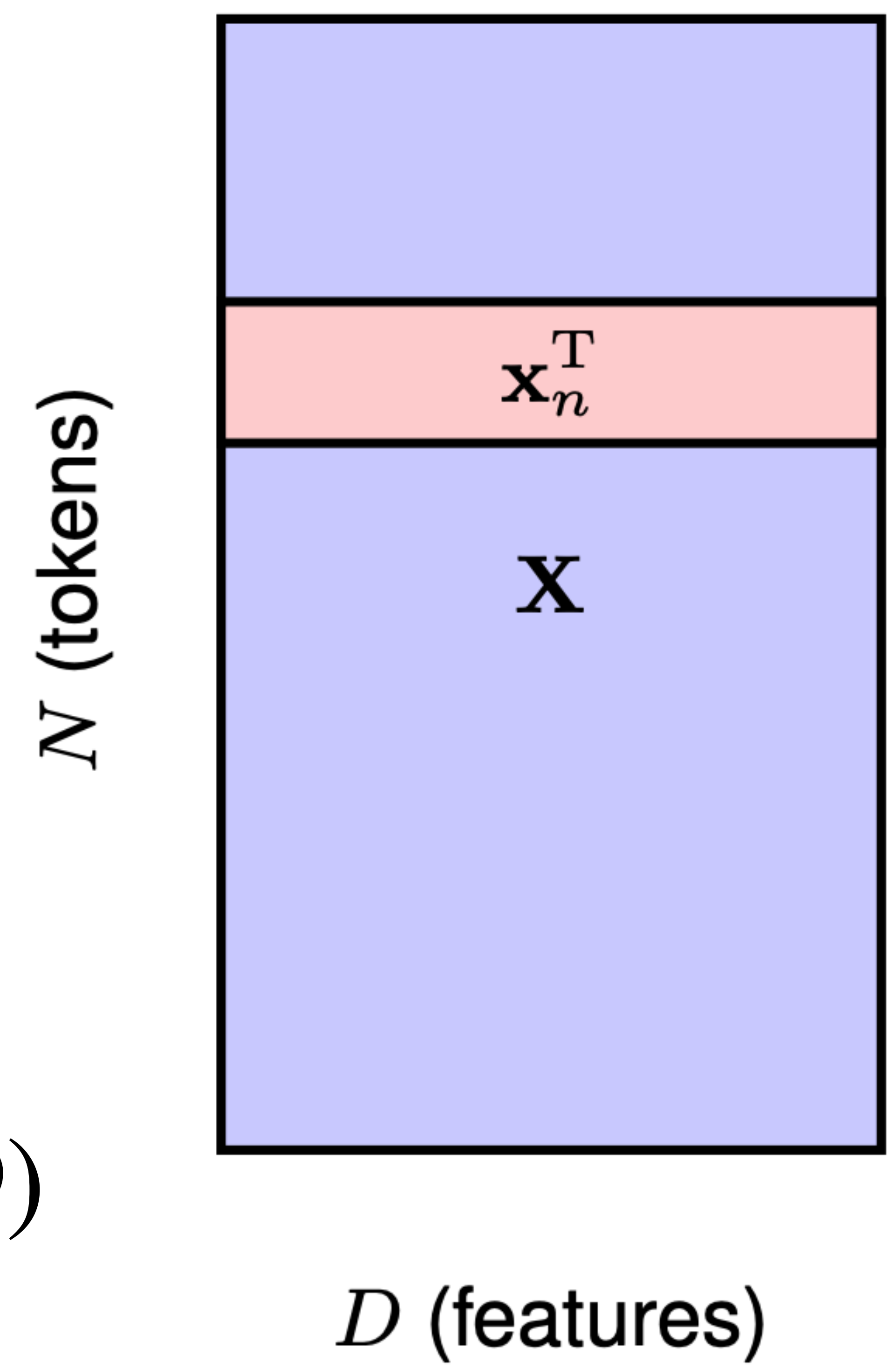
Tokenize building block of your semantic world in D dimensions

Make with N numbers of these buildings blocks the simplest organs of your world

$$X : N \times D$$

Your training data consists of lots of X

Feed the network with input (B, N_B, D)



Transformer

$$\mathbf{Y} = \mathrm{Softmax}\left[\mathbf{X}\mathbf{X}^{\mathrm{T}}\right]\mathbf{X}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 14 & 2 \\ 2 & 1 \end{bmatrix}$$

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$$= \begin{bmatrix} e^{0}/(e^{0}+e^{-12}) & e^{-12}/(e^{0}+e^{-12}) \\ e^{0}/(e^{0}+e^{-1}) & e^{-1}/(e^{0}+e^{-1}) \end{bmatrix}$$

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$$\mathrm{softmax}(XX^{\top}) \approx \begin{bmatrix} 0.999994 & 0.000006 \\ 0.731 & 0.269 \end{bmatrix}$$

Transformer

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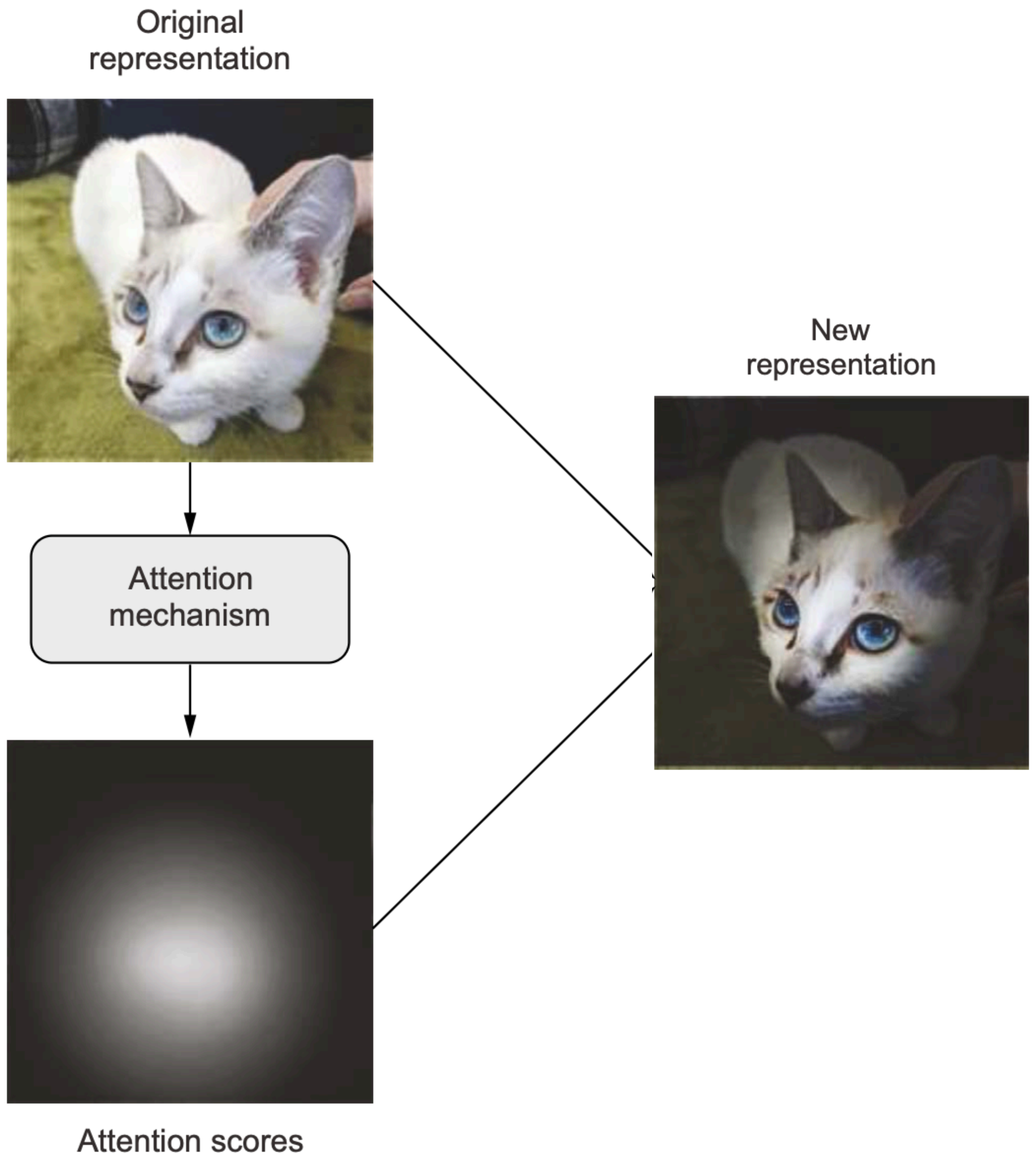
$$Y = \begin{bmatrix} 0.999994 & 2.000000 & 2.999982 \\ 0.731 & 1.731 & 2.193 \end{bmatrix}$$

Attention Mechanism

"Attention is all you need" by Vaswani et al.



Computing importance scores for a set of features, higher scores for more relevant features and lower scores for less relevant ones

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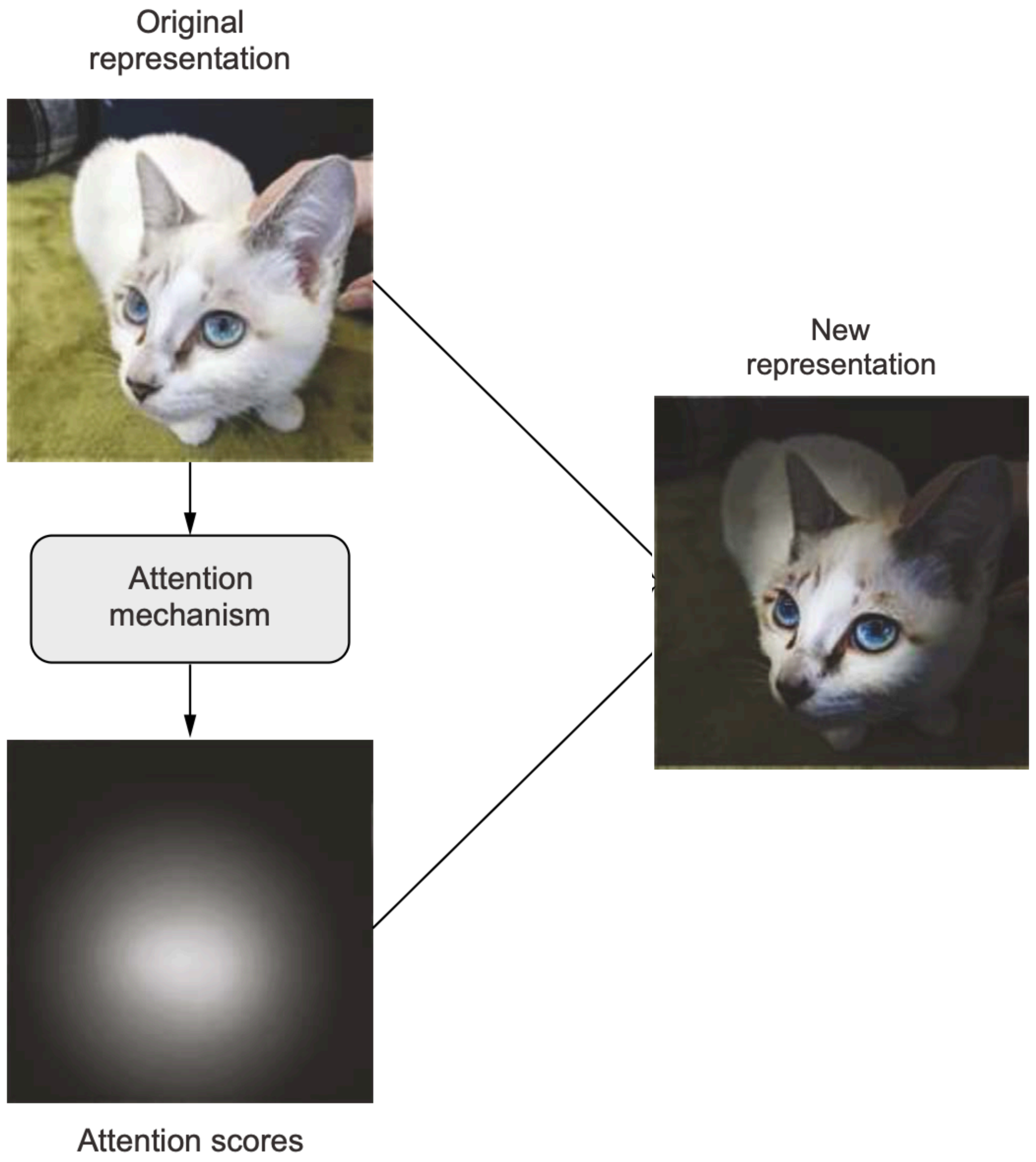


Computing importance scores for a set of features, higher scores for more relevant features and lower scores for less relevant ones

What is the criteria for the relevancy?

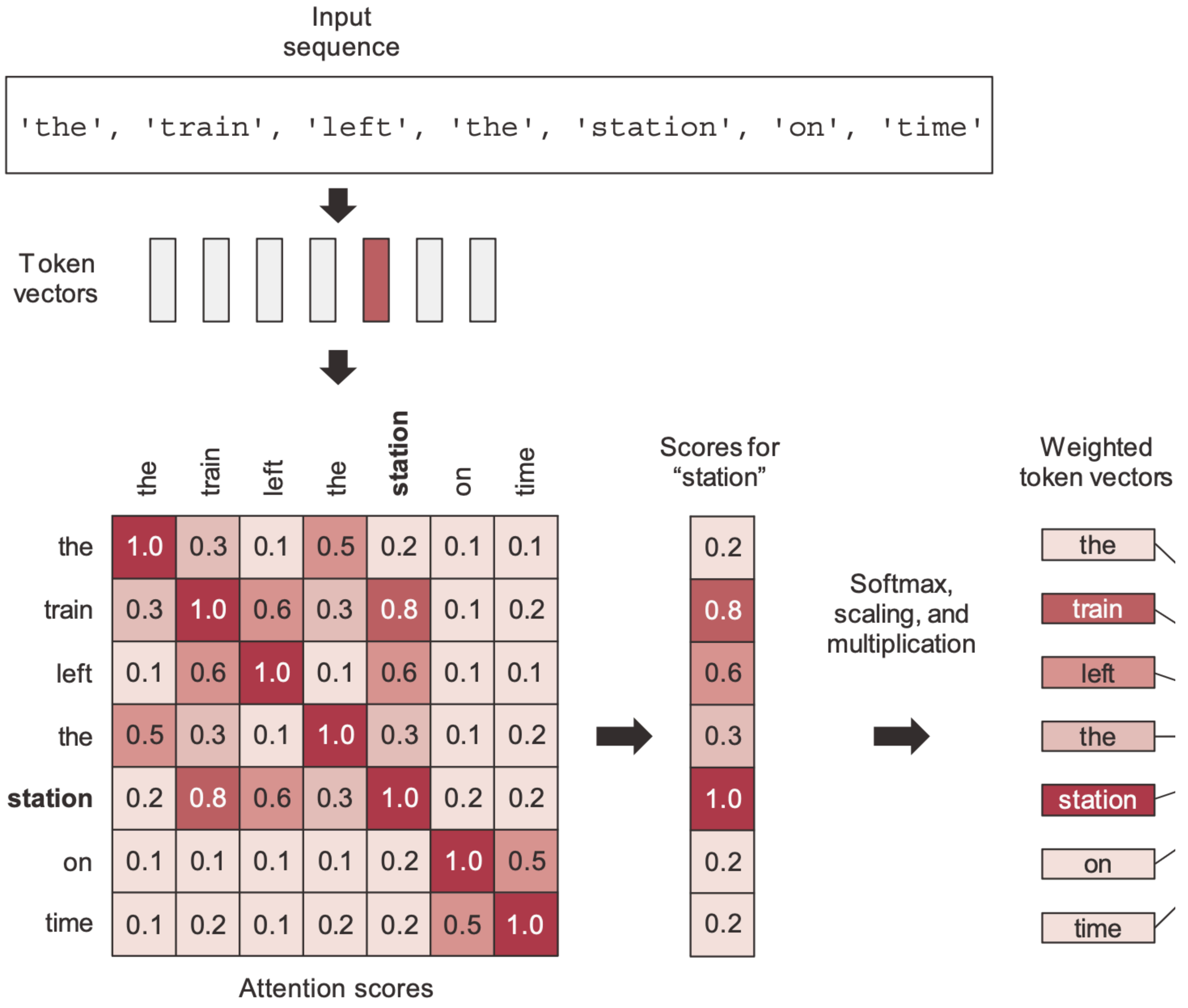
Self-Supervised can be scale!

Attention for languages

smart embedding space

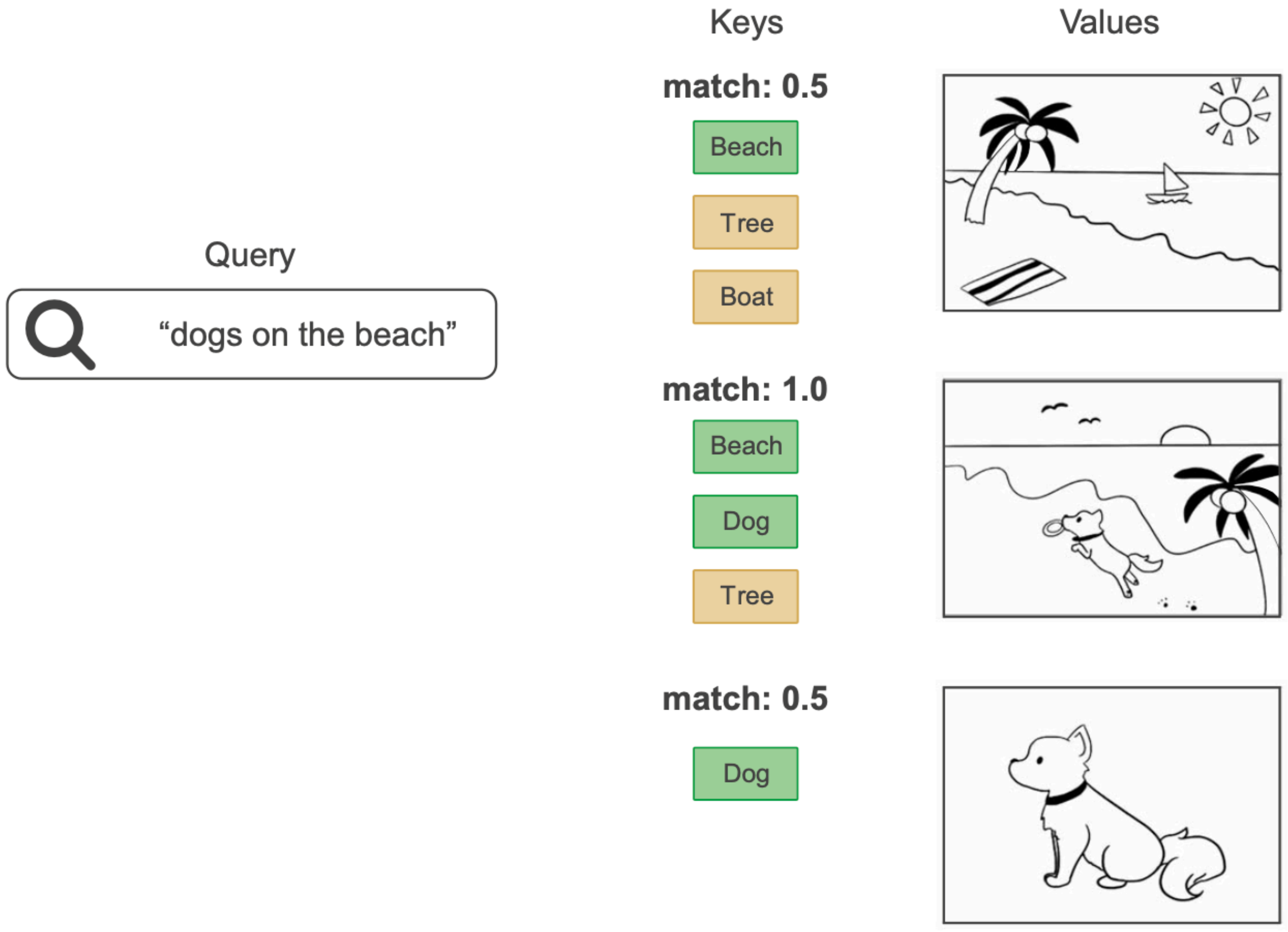
Context aware mechanism

modulate the representation of a token by using the representations of related tokens in the sequence



Attention Mechanism

sequence-to-sequence model



convert one sequence into another

$$\mathbf{Y} = \mathrm{Softmax}\left[\mathbf{Q}\mathbf{K}^{\mathrm{T}}\right]\mathbf{V}$$

$$\mathbf{Q} = \mathbf{X}\mathbf{W}^{(\mathrm{q})}$$
$$\mathbf{K} = \mathbf{X}\mathbf{W}^{(\mathrm{k})}$$
$$\mathbf{V} = \mathbf{X}\mathbf{W}^{(\mathrm{v})}$$

Large Language Model

Type of Foundation Model trained on language

e.g. BERT, GPT

The most general setup

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The most general setup

Number of independents basis for the semantic representation $\sim 50{,}000$

Overall sample is made of

100,000 books

1000 pages

10 lines

n words

$\sim 10^9$ block consist of n tokens

$$p(x_1, x_2, \ldots, x_n)$$

Number of free parameters 10^{11}

The domain is less than the complete sample $\sim 10^7$

Representative and Generative

Self-supervision at scale

BERT (Bidirectional Encoder Representations from Transformers)

Encoder

$p(x_i \mid x_2, \ldots, x_n)$

Reads the entire sentence at once → bidirectional representation.

Trained with Masked Language Modeling (MLM) and Next Sentence Prediction

GPT (Generative Pre-trained Transformer)

Decoder

$p(x_n \mid x_1, x_2, \ldots, x_{n-1})$

Looks only at past tokens → **left-to-right**, autoregressive.

Trained with next-token prediction

Foundation Models Concept

AI is undergoing a paradigm shift

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Machine is doing a task despite not being trained explicitly for it

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Emergence

Implicitly induced rather than explicitly constructed

Homogenization

Similar models and architecture, multi modal domains

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Training on broad data

Using Large size with respect to the population

Self-supervision at scale

Adapted to a wide range of downstream tasks

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https://crfm.stanford.edu/

https://arxiv.org/pdf/2108.07258